The effect of female role models on STEM major choice

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Abstract

This research analyzes how female role models (professors, teaching assistants and classmates), can affect major choice among female Civil Engineering students at the University of Chile. Using a conditional logit model, I estimate how exposure to female role models in the years prior to the decision influences the probability of choosing each academic major. Results show that female students attending mathematics or computer science courses taught by female professors are more likely to choose the mathematics and computer science majors, respectively. This effect is only observed for professors and not teaching assistants or classmates. Counterfactual analysis shows that increasing the number of female professors within majors where women are underrepresented, may encourage them to join these majors. The results also suggest that strengthening women's mathematics preparation may have comparable effects.

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1. Introduction

In today's society, the persistent gender wage gap remains a critical challenge that requires concrete and effective measures for its reduction. Data from the Chilean National Institute of Statistics (INE) show that in 2022 the gender gap in average income in Chile stood at 25,5 % to the detriment of women (INE, 2022).

Choosing a professional career is an important aspect that deserves careful analysis, as it directly influences occupational segregation between women and men. In Chile, three of the fields with the highest female participation, namely, Elementary Education, Early Childhood Education, and Nursing, have an average salary of $\$814,315$ Chilean pesos[2] in the second year after graduation. In contrast, fields with a high percentage of men, e.g., Electrical Civil Engineering, Electronic Civil Engineering, and Computer and Informatics Engineering, have an average income of $\$1,408,522$ Chilean pesos in the second year after graduation (MinMujeryEG, 2019).

In general, women tend to choose fields more closely related to the social sciences and the arts, whereas men prefer careers related to mathematics, computer science, or engineering, which are part of the so-called STEM fields[3]. These latter fields are characterized by a strong focus on mathematics and science, they tend to offer higher salaries, and are areas in which women are underrepresented. In Chile, according to data from the Higher Education Information System (SIES), one out of every four enrollments in STEM programs are of women. This gender gap has remained unchanged over the last five years (Subsecretaría de Educación Superior, 2023).

A relevant question to ask is why are women not enrolling in these fields despite the higher future returns they offer. Studies show that women do not choose higher-return majors because, when selecting a field of study, non-pecuniary preferences matter more than pecuniary ones (Turner & Bowen, 1999). Enjoying one's future work or obtaining parental approval are more important determinants, explaining half of male students' choices and more than 75 % of female students' choices (Zafar, 2013).

Although approximately one-third of the gender wage gap is attributed to differences in the occupations chosen by women and men, studies have shown that two-thirds of this disparity persists even within the same occupation (Goldin, 2015).

This research studies a specific group of future professionals who have already chosen a STEM degree and among whom gender gaps can be found in the different specializations they may select. In the Civil Engineering program at the University of Chile, all students enter a common curriculum for two years and then must choose a major or specialization[4] to complete their degree and graduate as Civil Engineers in Mathematics, Civil Engineers in Chemistry, Civil Engineers in Electricity, or in other specializations.

Research on individual decision-making highlights that, in addition to preferences and beliefs, so-

[2] Approximately 800 US dollars.

[3] For its initials in English (Science, Technology, Engineering, and Mathematics), groups three fields of knowledge according to ISCED-F-2013 (OECD): including Natural Sciences, Mathematics and Statistics; Engineering, Industry, and Construction; and Information and Communication Technologies.

[4] Hereafter used interchangeably.

cial interactions play an important role in shaping choices. In adittion, individuals' preferences can change, with context and social interactions acting as sources of learning and experience (Grotevant, 1987). These mechanisms are particularly relevant in educational settings, where interactions with instructors and peers can influence how students form expectations and evaluate different fields of study.

Role models that female students may have encountered before choosing their degree, as well as their peers, can be determinants in their choice of specialization and can play an especially important role in fields where women are underrepresented.

Much of the existing research focuses on the effect of role models (in particular professors) on students' academic performance, while fewer studies analyze how these role models influence students' choice of major. In general, findings show that female professors have a positive effect on female students: higher exposure to female professors in a department increases the proportion of female students who choose majors related to that department (Fox, 1974; Rask & Bailey, 2002; Bettinger & Long, 2005).

On the other hand, regarding the effect that peers (female classmates and students from older cohorts) may have on the choice of major, it has been found that greater interaction with students of a particular gender increases the likelihood that students will follow that gender's career-choice patterns in university (Bordón, Canals & Mizala, 2020; Brenøe & Zölitz, 2020).

The objective of this research is to analyze how social interactions, particularly the exposure to female role models during the first two years of the program, affect the specialization decisions of Civil Engineering female students at the University of Chile. The group of female students under study has already entered a STEM degree; they can be considered to have overcome one barrier by choosing a field in which women are underrepresented. I explore whether, once inside the program, their preferences for one specialization over another can be influenced by female role models in their academic environment, such as professors and teaching assistants (TAs) acting as role models, or by female classmates and women from previous cohorts who may exert peer effects.

To address this question, specialization choice is modeled using a random utility framework (McFadden, 1974). The indirect utility each student obtains from choosing a specialization depends both on characteristics specific to each field and on individual attributes. Using data from the first two years of coursework for Civil Engineering students, the parameters of interest are estimated through a conditional logit model to evaluate whether female role models affect students' specialization choices.

This research differs from the existing literature in that it does not limit its focus to the influence of professors; it also examines the role of female teaching assistants, female students from previous cohorts within each specialization, and female classmates from the same cohort.

The findings of this research support the positive effect that female professors have as role models on female students' specialization choice. However, teaching assistants are found to have the opposite effect, discouraging students from entering a specialization in which they were taught by a larger number of female teaching assistants.

The counterfactual analyses conducted offer interesting insights: increasing the share of female professors teaching common-curriculum courses slightly increases the participation of women in specializations where they are underrepresented. Additionally, to encourage more female students to enter Computer Science and Mathematics specializations, Mathematics Standardized College Entrance Exam (PSU) scores have a strong effect on their likelihood of entering those fields.

The document is organized as follows: after this introduction, Section 2 presents a review of the related literature. Section 3 details data and methodology used to estimate the model. Section 4 presents results, together with the counterfactual exercises and their discussion. Finally, Section 5 concludes by outlining the study's limitations and the future research it motivates.

2. Literature Review

The choice of major for future professionals is an important topic that can help explain the gender wage gap. The role models that female students may have encountered before choosing their degree, as well as peer effects, can be determinants of their decisions when choosing a major or specialization.

The role-model effect can be associated with different definitions. Morgenroth et al. (2015) suggest that the concept can be summarized into three dimensions: role models show us how to perform a skill and reach a goal (behavioral models), they show us that a goal is attainable (representations of what is possible), and they make a goal desirable (sources of inspiration).

When studying female students' choice of major and how to motivate them to enter disciplines in which they are underrepresented, it is important to make the goal both attainable and desirable. In this context, different possible role models can be identified, from mothers and fathers, to school and university teachers, to public figures, who may act as behavioral models and representations of what is possible (for example, pursuing a STEM career), but who can also serve as inspiration, encouraging students to want to be part of that group.

Research shows that career choices are often guided by parental influences, and the effects differ between fathers and mothers. Sons tend to be influenced by their fathers, whereas daughters are influenced by the occupations of both parents. Moreover, mothers have stronger effects on their daughters in specific areas such as health, business, social sciences, humanities, and law (Bordón, Canals & Mizala, 2020). Sikora & Pokropek (2012a) using data from several countries, find that in many of them, fathers' occupations tend to influence sons more, whereas in relatively few countries mothers' occupations shape daughters, and even then, only in certain fields such as biology, agriculture, and health-related careers.

Some policies and efforts aimed at increasing the representation of women in male-dominated disciplines[5] focus on expanding the orientation opportunities available to female students, partly by hiring more female professors. These professors can play an important role in encouraging female students to enroll in programs where they are underrepresented (Bettinger & Long, 2005; Porter

[5]Defined as fields in which the majority of practitioners are men.

& Serra, 2020).

The Civil Engineering program at the University of Chile has several specializations. Students choose their track after two years of common-curriculum courses, and the distribution of men and women across specializations is not homogeneous, for example, in the Mathematics specialization women are underrepresented.

If there is indeed a role-model effect from female professors, one would expect that exposure to them during the first two years would encourage female students to choose a particular major, possibly one in which they are underrepresented, since female professors may serve as examples and relate more closely to students' needs.

Much of the literature studying the role-model effect of professors focuses on how they may influence students' academic performance, since perceiving professors as role models can increase students' interest and engagement in the subject (Hoffmann & Oreopoulos, 2009).

At the school level, studies show that students' academic performance may improve when courses are taught by a teacher of the same gender (Dee, 2007). In contrast, Cho (2012), using data from 15 OECD countries, reported results contradicting this hypothesis. Some argue that the positive performance effect when teacher and student share the same gender may in fact be driven by a negative effect generated when the teacher is of the opposite gender. Paredes (2013) studies eighth-grade female students in Chile and concludes that there is a positive role-model effect of female teachers on girls' academic performance, without any negative effect on male students.

At the university level, Hoffmann & Oreopoulos (2009) find a small positive effect, driven by male students performing worse when taught by female professors, while female students respond similarly to professors of both genders. Carrell et al. (2010), using a sample of university students who are randomly assigned to instructors, find a positive effect on performance in mathematics and science courses. The effect is small for male students but particularly large for female students (and even stronger for high-achieving women in mathematics). They also find a positive effect of instructor gender on the probability of graduating with a STEM degree, consistent with the findings of Neumark & Gardecki (1998) for PhD students in economics.

Although a large body of research examines the effects of role models on academic performance, fewer studies focus on their effect on students' choice of major. The literature has approached role models by considering variables such as the number of female professors in each department associated with degree programs, or the professors with whom students take classes when entering university. If the first years of university are a formative experience that may affect future decisions about major or specialization, the professors students encounter in early courses may influence the choices they later make.

Fox (1974) is one of the first to study a positive correlation between the number of female professors and the number of female students in a major across several universities. Although he does not analyze variation over time, he interprets his results as supporting the existence of a same-gender role-model effect in the choice of major.

Subsequent research incorporates year-to-year variation by examining the proportion of female faculty in each department and the proportion of female students who choose majors associated with that department. Canes & Rosen (1995), studying students at Princeton University, the University of Michigan, and Whittier College, find no effect of increasing the share of women faculty in a department on the percentage of female students who graduate from that department.

Unlike previous studies, Bettinger & Long (2005) and Rask & Bailey (2002) examine students' exposure to female professors rather than the share of women in a department. Using the percentage of female professors who taught female students in courses associated with each major, they found a positive effect.

Bettinger & Long show that this positive effect is particularly strong for female students in mathematics and statistics majors, but they do not find a significant effect in fields such as engineering, physics, and computer science. When studying male role models, they find a positive effect in majors such as education, where men are underrepresented.

In addition, the effect of female teaching assistants on female students' choice of major has also been examined. Griffith & Main (2021) find a positive effect of female teaching assistants on female students' choice of specialization.

Another variable to consider when studying major choice is the peer effect, which classmates or cohort peers may have on students' behaviors or decisions. A peer effect may be considered when, given a student's educational resources, a classmate influences an educational outcome of that student. This effect can be direct, where the presence of one student affects the outcome of another without either changing their behavior, or indirect, where a classmate's specific attitude motivates the student to adjust their behavior and follow that of their peers (Epple & Romano, 2011).

Part of the related literature studies the classmates that students interact with in their final years of school and evaluates whether their gender has any effect on the majors students later choose. It has been found that greater interaction with students of a particular gender in school increases the likelihood that students will follow that gender's patterns of major choice in university. Bordón, Canals & Mizala (2020) show this using Chilean data for a wide variety of majors, while Brenøe & Zölitz (2020) find the same for STEM majors, reporting a negative effect on the probability that female students choose STEM fields when they have more female classmates. Conversely, Favara (2012) shows that in single-sex schools, girls tend to make less stereotypical career choices, and Ashworth & Evans (2001) also find a positive effect of female classmates increasing the probability that female students choose economics, a male-dominated field.

At the university level, there is less evidence on the effects that peers from the first years of college may have on major or specialization choice. Zölitz & Feld (2020) work with same-course classmates and find that having more female peers reinforces gender-stereotypical patterns of specialization choice, similar to the findings of Bordón, Canals & Mizala (2020) and Brenøe & Zölitz (2020).

Rask & Bailey (2002), when studying older cohorts of female students, find a positive effect: a higher percentage of women in the previous cohort of a specialization increases the likelihood that a female student will choose that specialization.

There are other gender-related characteristics studied in the major-choice literature, such as risk aversion, self-confidence, and attitudes toward competition. Most evidence shows that women are more risk-averse than men and more likely to avoid competitive environments (Horn & Kiss, 2019; Byrnes et al., 1999). Bordón, Canals & Mizala (2020) find with Chilean data that male students tend to choose more selective majors than their female counterparts when they have high PSU scores. In mixed and more competitive environments, girls tend to have less successful outcomes than male peers with the same characteristics (Gneezy et al., 2009). Moreover, girls' self-perceptions and confidence in their mathematical and scientific abilities are lower than those of boys (Sikora & Pokropek, 2012b), which means that women require stronger signals of mathematical ability to decide to pursue a male-dominated major (Justman & Méndez, 2018).

3. Data and Methodology

3.1. Data

The Civil Engineering program at the University of Chile has a common curriculum during the first two years, which includes general training courses for the degree as well as introductory courses to certain specializations. Starting in the third year, students must enroll in a specialization and take the corresponding courses, following a curricular track with specific requirements in order to obtain the degree of Civil Engineer in the chosen specialization. Among the specializations students may choose are Civil Engineering: Mathematics, Electrical, Mechanical, Industrial, Chemical, Mining, Biotechnology, Computer Science, Transportation, Hydraulic, Sanitary and Environmental Engineering, and Structures and Construction.

Upon entering the common curriculum of Civil Engineering, students may also choose, in addition to the previously mentioned options, one of four science bachelor's programs with a concentration in Physics, Astronomy, Geophysics, or Geology. Students who opt for a bachelor's degree constitute a small and distinct group, as most of the time they do not enroll in the common curriculum with the goal of obtaining an engineering degree.

During the first semester, students are randomly assigned to a group that shares the same sections in all of their courses, functioning similarly to a school class, as they will meet in each course throughout that semester. From the second semester onward, students may freely choose the sections and courses they take, following the structure of the common curriculum.

This study uses data from several sources: the Chilean Ministry of Education, the Academic Management Office of the Faculty of Physical and Mathematical Sciences (FCFM) of the University of Chile, and the Faculty's Office of Diversity and Gender. The dataset includes students who entered Civil Engineering at the University of Chile between 2008 and 2018 and contains their PSU scores[6] in Mathematics and Verbal, type of university admission, gender, cohort, chosen specialization, and current academic status (graduate, degree holder, or active student). Moreover, the available data allow identification of the courses and sections taken each semester, the instructors teaching them,

[6]PSU: Prueba de Selección Universitaria, Chile's standardized college entrance exam, containing two mandatory tests: Mathematics and Verbal, and two optional tests: Science and History.

and the grades students obtained in their early years of study[7].

The sample includes students who entered the program between 2009 and 2018, who enrolled in a specialization and are therefore pursuing it, or who have already graduated with their corresponding degree. The sample includes students for whom gender information is available and for whom it was possible to identify the course section they were assigned to during the first semester[8].

The Faculty of Physical and Mathematical Sciences allows its students to obtain more than one specialization. However, to study specialization choice and the effects of female role models more cleanly, only students who declared and/or completed one specialization are included. Additionally, not all previously mentioned specializations are considered due to differences in student profiles and motivations, especially for those who choose science bachelor's programs rather than an engineering degree. The final sample includes students who chose among the following specializations: Mathematics, Electrical, Mechanical, Industrial, Chemical, Mining, Biotechnology, and Computer Science, resulting in a sample of 897 women and $3,231$ men.

The sample of instructors includes both female professors (lecturers) and female teaching assistants[9], restricted to those who taught courses in the common curriculum during the years for which information is available. This results in a total of 2,058 instructors for whom it is possible to identify the department offering each course, for each semester and year.

The common curriculum courses that students must complete during the first two years are summarized in Table 1[10]. These courses are essential for linking the instructors who teach them to the specializations students may later choose. In the Faculty, courses with prefixes (MA) are taught by the Mathematics Department, (FI) by Physics, and so on: Computer Science (CC), Industrial Engineering (IN), Chemistry (CM). Other courses correspond to introductory or interdisciplinary engineering classes (EI–CD), and are not tied to a single department.

It is important to note that students' exposure to each specialization during the first two years is not uniform, since only some specializations have related courses in the common curriculum.

Table 1: Common curriculum courses.

1° Año		**2° Año**	
I semestre	II semestre	I semestre	II semestre
MA1001	MA1002	MA2001	IN2201
MA1101	MA1102	MA2601	MA2002
FI1000	FI1100	FI2003	FI2002
FI1001	CD1201	FI2001	FI2004
CC1000	CC1002	CM1001	CM2004
EI1101	EI1102	EI2001	CD2201

[7] My dataset includes only the courses a student has passed, not those they have failed.

[8] Allowing them to be matched with their first-semester classmates.

[9] Teaching assistants are generally senior students or recent graduates who have completed the course and assist the instructor by teaching the practical sessions.

[10] Although academic programs change over the years, these courses remain part of the curriculum and are used in this study

3.2. Methodology

Model

The problem faced by second-year students in the common curriculum is deciding which specialization they will choose to complete their engineering degree. To study this choice problem, I use a random utility model[11], where a student's utility from choosing a specialization is a linear function of characteristics of the specialization and characteristics of the student.

Each student i aims to maximize her utility u_i. Therefore, if the utility of choosing specialization j, u_{ij}, is greater than the utility of choosing any other available specialization, she will choose j. The probability that student i chooses specialization j is given by the following expression[12], where y_i is a variable indicating the specialization chosen by the student.

$$p_{ij} = Pr(y_i = j) = Pr(u_{ij} > u_{ik}) \quad \forall k \neq j \tag{1}$$

The probability of choosing one specialization over another may depend both on characteristics specific to each alternative and on characteristics specific to each student. I estimate major choice using a conditional logit model, which includes variables specific to alternatives (specializations) and to individuals (students). The indirect utility of student i from choosing specialization j is defined as:

$$u_{ij} = a_j + x'_{ij}\beta + z'_i\gamma_j + \lambda_g + \epsilon_{ij} \tag{2}$$

Here, x'_{ij} corresponds to characteristics of each alternative, which may also vary at the student level. Four alternative-specific variables are considered relevant in this analysis, the percentage of female professors, the percentage of female teaching assistants, the percentage of female classmates in common-curriculum courses related to specialization j, and the percentage of female students in the previous cohort enrolled in that specialization.

A single parameter β measures the effect of each of these xxx variables on the student's indirect utility, meaning that a one–percentage point increase in female professors/assistants/classmates is assumed to have the same marginal (dis)utility across all specializations.

I also include an indicator variable, *no_plancomun*, which identifies specializations that do not have courses taught in the common curriculum. This allows utility from choosing a specialization to depend on the exposure the student had during the first years of the program to related subjects.

It is important to note that the Mechanical, Mining, Biotechnology, and Electrical specializations do not have any common-curriculum courses in any cohort. By contrast, Computer Science has common-curriculum courses only for some cohorts. For cohorts 2009–2013, the curriculum does not include Computer Science courses, and these cases are also captured by this indicator.

On the other hand, z'_i includes student-specific variables that do not vary across alternatives. These are the students' PSU Mathematics and PSU Verbal scores. These variables capture differences in preferences for each specialization as a function of PSU scores, and thus the coefficients γ_j for each

[11] See McFadden (1974)

[12] Note that the model does not allow two majors to have identical utility. As noted above, even though students are allowed to complete more than one major, the sample includes only those who selected one.

alternative are of interest.

Finally, to control for potential changes across years or cohorts in preferences or in the popularity of certain specializations, I include a cohort fixed effect,λ_g, which does not vary across students or specializations. The error term, ϵ_{ij} , follows a Type I Extreme Value distribution, representing unobserved shocks affecting utility from choosing a specialization. Under these assumptions, the probability that student i chooses specialization j is:

$$p_{ij} = Pr(y_i = j) = \frac{e^{a_j + x'_{ij}\beta + z'_i\gamma_j + \lambda_g}}{\sum_{l=1}^{J} e^{a_l + x'_{il}\beta + z'_i\gamma_l + \lambda_g}} \quad , \quad l = 1, 2, ..., J \tag{3}$$

Model Identification

The model evaluates different types of female role models that may influence students' decisions. The percentage of female professors and female teaching assistants measures the effect of role models, while the proportion of female classmates and female students in the previous cohort is included to capture peer effects, as these may influence the behavior of students and motivate them to choose one specialization over another.

Estimating these effects may face selection bias issues. Civil Engineering students at the University of Chile are randomly assigned to their courses only during the first semester; beginning in their second semester, students may choose the instructor with whom they take each course.

If female role models exert influence through their early contact with students, by teaching them or by being their classmates, it is possible that students with certain characteristics may prefer to select sections with a higher (or lower) number of female professors or female classmates, depending on their preferences. Even if a student already has a strong preference for a particular specialization, she may have an even stronger preference for taking a course with a female professor associated with that specialization. For this reason, to correctly identify the coefficients of interest, the percentages of female professors/assistants/classmates must be exogenous to the student's specialization choice.

Previous studies analyzing the effect of role models on career choice use as explanatory variables the percentage of female faculty in a department during the years prior to specialization choice (Canes & Rosen, 1995). Since the composition of the faculty in a given year is not something students can anticipate or expect to change, this variable is exogenous to students' specialization decisions. However, the authors do not assume that students actually take courses with those faculty members, which may make it more difficult for female instructors to influence students to choose that department's specialization.

Studies such as Rask & Bailey (2002) and Bettinger & Long (2005) identify the match instructor-student and use as their variable of interest the proportion of classes, related to the specialization, that students take with female professors. Rask & Bailey do not account for this selection bias in their estimation, whereas Bettinger & Long address the problem using an instrumental variable that captures deviations in each period from the steady-state share of courses taught by women.

Studying how the percentage of female faculty in a department affects students' choices may reveal the environment they would face if they were to join that department and the expectations this

generates regarding the specialization. By contrast, studying the percentage of female professors with whom a student actually took classes captures the level of exposure and direct contact, and how this may encourage students to enter a particular specialization. For this research, I focus on how prior exposure to female role models, before the specialization decision, is relevant for shaping students' choices.

To identify the causal effect, I use the predicted percentage of female professors teaching each common-curriculum course, a variable that indicates how likely it is for a student to take a course with a female instructor, which is exogenous to the student's specialization choice. To construct this measure, I exploit the random assignment of students to first-semester courses and the variation over time in the percentage of female professors teaching each course across cohorts.

Students may manipulate course timing by delaying their courses to semesters with a higher number of female instructors. To prevent this bias, the construction of the variables (percentage of female professors, teaching assistants and classmates) uses for each cohort the common-curriculum courses scheduled in the semester corresponding to the official curriculum.

Construction of the Variables of Interest

The variables used to evaluate the effects of role models and peers include: the percentage of female professors, female teaching assistants, female classmates, and female students from the previous cohort. The first three variables are constructed in the same way: for each common-curriculum course taken by a student, the predicted percentage of female professors/TAs/classmates is calculated.

To exploit the random assignment of students in the first semester, the groups of students assigned to the same course sections during that semester were identified. For each group, the proportion of female professors, teaching assistants, and classmates with whom they took each course was computed.

For the remaining semesters of the common curriculum, the percentage of female professors, assistants, and classmates is first calculated for each section of each course, separately by semester. Once the percentage of female professors in each section of course *CC1002*, for example, is obtained, the average percentage across all sections is computed, yielding a measure of how likely it is that a student takes that course with a female professor.

After repeating this procedure for each course in semesters *II*, *III*, and *IV*, together with the information from the first semester, the three variables (female professors, assistants, and classmates) are obtained for every course in each semester. The final values for these variables are computed by averaging across all courses related to each specialization that are offered during the four semesters of the common curriculum[13].

The last variable of interest is the percentage of female students from the previous cohort who chose each specialization. This variable is constructed by dividing the number of women in the previous

[13]These variables were constructed so that having two women in the same section/course does not artificially increase the probability of taking a course with a female instructor. Percentages were calculated first at the section level, then at the course level, and finally at the department/specialization level.

cohort who enrolled in a specialization by the total number of students in that specialization.

Table 2 presents the mean and variation of the variables of interest by cohort and specialization. For the first three variables, only specializations with common-curriculum courses are shown, since data for these variables are not available for the remaining specializations. When these variables are missing, as in the case of Computer Science between 2009 and 2013,they are replaced with zero but are identified with the model's *no_plancomun* variable. The statistics for the variable capturing female students in the previous cohort are shown for all specializations.

Table 2: Variables' distribution by cohort.

	2009		2010		2011		2012		2013		2014		2015		2016		2017		2018	
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.
% Profesoras																				
Computación	.	.	.	.	.	.	.	.	.	.	0.0	0.0	18.4	21.5	12.3	21.6	17.6	21.7	12.7	21.8
Química	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	5.0	0.0	5.6	0.0	12.5	0.0	12.5	0.0	18.8	0.0	50.0	0.0
Industrial	25.0	0.0	33.3	0.0	33.3	0.0	33.3	0.0	16.7	0.0	33.3	0.0	33.3	0.0	33.3	0.0	33.3	0.0	33.3	0.0
Matemáticas	7.0	5.1	4.3	4.8	6.5	4.7	9.7	4.7	10.2	6.9	12.1	6.2	12.3	9.7	10.2	10.4	5.8	6.3	8.6	6.3
% Ayudantes																				
Computación	.	.	.	.	.	.	.	.	.	.	21.7	18.5	24.6	24.1	28.4	18.1	19.0	17.1	21.9	10.6
Química	33.3	0.0	0.0	0.0	33.3	0.0	37.5	0.0	10.1	0.0	7.3	0.0	30.3	0.0	31.3	0.0	38.2	0.0	12.5	0.0
Industrial	0.0	0.0	36.1	0.0	77.8	0.0	11.1	0.0	23.1	0.0	22.4	0.0	17.4	0.0	11.1	0.0	11.1	0.0	77.8	0.0
Matemáticas	6.1	2.6	6.7	4.9	3.6	5.0	5.3	4.8	13.7	4.6	13.8	7.1	9.7	6.3	19.6	6.9	14.0	6.0	9.7	6.8
% Compañeras																				
Computación	.	.	.	.	.	.	.	.	.	.	23.7	3.1	22.3	2.2	24.8	1.8	27.2	3.2	27.8	3.7
Química	28.3	0.0	26.4	0.0	20.6	0.0	26.8	0.0	22.6	0.0	27.0	0.0	29.6	0.0	31.0	0.0	30.0	0.0	28.2	0.0
Industrial	21.2	0.0	15.7	0.0	20.3	0.0	23.3	0.0	20.7	0.0	34.3	0.0	35.2	0.0	31.3	0.0	26.4	0.0	30.4	0.0
Matemáticas	16.7	1.3	15.7	1.2	14.2	1.6	18.4	1.4	15.8	1.2	21.4	1.8	20.1	1.3	22.6	1.0	25.7	1.8	24.8	2.1
% Gen anterior																				
Computación	11.5	0.0	15.2	0.0	9.3	0.0	18.4	0.0	12.1	0.0	8.9	0.0	15.6	0.0	22.5	0.0	15.6	0.0	22.7	0.0
Industrial	27.9	0.0	27.9	0.0	20.1	0.0	18.3	0.0	23.5	0.0	24.2	0.0	34.1	0.0	34.2	0.0	39.5	0.0	35.0	0.0
Eléctrica	5.0	0.0	5.5	0.0	10.0	0.0	4.9	0.0	22.6	0.0	8.6	0.0	14.6	0.0	14.5	0.0	13.0	0.0	16.5	0.0
Biotecnología	75.0	0.0	50.0	0.0	40.0	0.0	75.0	0.0	0.0	0.0	42.9	0.0	55.6	0.0	50.0	0.0	50.0	0.0	52.6	0.0
Matemáticas	0.0	0.0	5.9	0.0	11.1	0.0	9.1	0.0	15.0	0.0	0.0	0.0	0.0	0.0	8.0	0.0	9.7	0.0	18.8	0.0
Química	22.2	0.0	19.0	0.0	35.3	0.0	28.6	0.0	36.8	0.0	28.0	0.0	37.5	0.0	42.3	0.0	51.5	0.0	43.5	0.0
Mecánica	7.5	0.0	23.8	0.0	4.9	0.0	8.0	0.0	11.6	0.0	18.5	0.0	23.4	0.0	11.1	0.0	26.1	0.0	18.3	0.0
Minería	21.7	0.0	17.4	0.0	6.9	0.0	19.1	0.0	17.6	0.0	14.3	0.0	22.7	0.0	7.7	0.0	12.5	0.0	66.7	0.0
Observaciones	360		390		373		391		422		448		460		459		438		387	

3.3. Descriptive Statistics

Figure 1 displays the proportion of female students who choose each specialization, with a dashed line indicating the overall average share of women in the sample. It is evident that in the Computer Science, Mathematics, Electrical, and Mechanical specializations, women are underrepresented, making up less than 22 % of students in these specializations across all cohorts. By contrast, the Industrial, Biotechnology, Chemistry, and Mining specializations have consistently had higher female participation over the years, with some showing increasing trends.

Figure 1: Female students' major distribution by cohort.

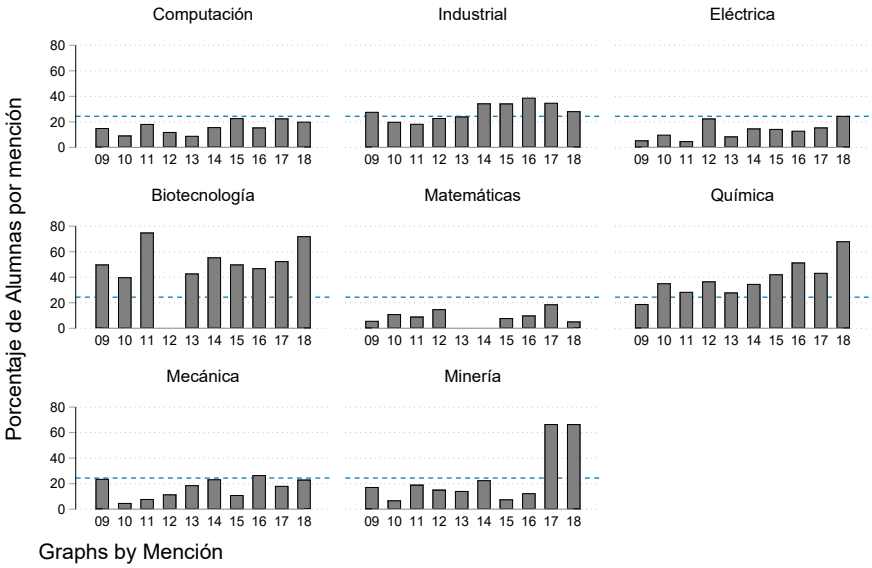


Table 3 provides descriptive statistics for the student sample. The vast majority of students (92,5 %) were admitted through the regular university admission process (PAA or PSU), but the sample also includes students admitted through the Academic Excellence Scholarship, athlete quota, the bachelor's program, and the Gender Equity Priority quota. The latter is a special admission track for women available since 2014, which currently allows the 85 women who remained on the wait list under regular admission criteria to enter the program (University of Chile, n.d.).

Female students have, on average, lower PSU Mathematics scores than male students, and they slightly outperform men on the PSU Verbal test. Overall scores in the sample are high, reflecting the program's high selectivity, as only students with very strong academic performance are admitted.

There are also notable gender differences in specialization choices. A small percentage of female students (2,5 %) choose the Mathematics specialization, compared with 6,8 % of male students. Similarly, 11,8 % of women choose Computer Science, compared to 15,8 % of men. Interestingly, the specialization where male students are most underrepresented is Biotechnology: only 1,4 % of male students enroll in this specialization, compared with 5,7 % of female students. The most popular specialization for both genders is Industrial Engineering, chosen by 43,1 % of women and 30 % of

men. However, men also show a strong preference for the Electrical specialization, with 21,3 % enrolling in it, while women tend to cluster in Industrial Engineering.

Table 3: Descriptive Statistics.

	Mujer		**Hombre**		**Total**	
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.
Tipo de Ingreso						
Beca Excelencia Académica - BEA	2.8	16.5	2.0	14.1	2.2	14.7
Egresados/as y titulados/as	0.0	0.0	0.0	1.8	0.0	1.6
Bachillerato	1.4	12.0	0.5	6.8	0.7	8.2
Deportista	1.0	10.0	1.1	10.6	1.1	10.5
Prioridad Equidad de Género	11.5	31.9	0.0	0.0	2.5	15.6
Ingresos Especiales	0.0	0.0	0.1	2.5	0.0	2.2
PAA o PSU	81.6	38.8	95.5	20.7	92.5	26.4
SIPEE	1.7	12.8	0.7	8.2	0.9	9.4
Transferencias	0.0	0.0	0.1	2.5	0.0	2.2
Puntajes PSU						
Puntaje PSU Matemáticas	747.1	43.4	778.8	43.1	771.9	45.1
Puntaje PSU Lenguaje	693.2	55.8	688.8	59.2	689.7	58.5
Menciones						
Computación	11.8	32.3	15.8	36.5	15.0	35.7
Industrial	43.0	49.5	30.0	45.8	32.8	47.0
Eléctrica	12.0	32.6	21.3	40.9	19.3	39.5
Biotecnología	5.7	23.2	1.4	11.7	2.3	15.1
Matemáticas	2.5	15.5	6.8	25.2	5.9	23.5
Química	10.1	30.2	4.3	20.3	5.6	22.9
Mecánica	9.9	29.9	13.7	34.4	12.9	33.5
Minería	4.9	21.6	6.7	25.0	6.3	24.3
Observaciones	897		3,231		4,128	

Finally, Table 4 presents the distribution of PSU scores for students in each specialization. First, it is important to note that all specializations include students with maximum scores on the Mathematics exam[14]. For the Verbal exam, average scores are not at the maximum but are around 700 points, which is a strong performance on the PSU.

Specializations with the highest PSU Mathematics averages are Electrical, Mathematics, and Mechanical Engineering, with mean scores above 775. Students in the Mathematics specialization have, on average, the highest PSU Mathematics scores, women have 781,5 points and men 801,7 points. In contrast, the Industrial, Chemistry, and Biotechnology specializations have lower PSU Mathematics averages, around 750 points.

[14]PSU scores range from 150 to 850 points, with 850 being the maximum score.

Table 4: PSU scores distribution by major.

	Mujer		**Hombre**		**Total**	
	Mate.	Leng.	Mate.	Leng.	Mate.	Leng.
Computación						
Mean	747.1	703.7	776.4	700.4	771.4	700.9
Std.Dev.	43.6	53.2	43.6	58.4	44.9	57.5
Min	648.0	579.0	673.0	542.0	648.0	542.0
P50	743.0	708.0	773.5	697.0	770.0	700.0
Max	850.0	814.0	850.0	831.0	850.0	831.0
Industrial						
Mean	743.3	692.4	774.2	683.7	765.4	686.2
Std.Dev.	41.6	54.3	42.8	58.8	44.6	57.7
Min	624.0	537.0	630.0	502.0	624.0	502.0
P50	743.0	692.0	772.0	682.0	763.0	684.0
Max	850.0	821.0	850.0	834.0	850.0	834.0
Eléctrica						
Mean	761.0	696.0	782.8	691.5	779.9	692.1
Std.Dev.	45.9	55.6	42.6	59.2	43.7	58.7
Min	630.0	539.0	623.0	508.0	623.0	508.0
P50	755.0	692.0	785.0	691.0	784.0	691.0
Max	850.0	850.0	850.0	850.0	850.0	850.0
Biotecnología						
Mean	734.5	702.2	765.5	703.9	749.0	703.0
Std.Dev.	41.5	61.3	45.9	52.1	46.1	56.9
Min	669.0	565.0	688.0	588.0	669.0	565.0
P50	727.0	705.0	762.0	692.0	745.0	695.5
Max	850.0	799.0	850.0	799.0	850.0	799.0
Matemáticas						
Mean	781.5	714.7	801.7	704.8	799.9	705.7
Std.Dev.	48.5	61.7	38.6	63.5	39.9	63.2
Min	681.0	616.0	690.0	466.0	681.0	466.0
P50	786.5	715.5	809.0	704.5	809.0	705.5
Max	850.0	802.0	850.0	834.0	850.0	834.0
Química						
Mean	745.1	685.1	763.4	686.1	756.1	685.7
Std.Dev.	42.2	59.3	41.7	56.6	42.8	57.6
Min	662.0	501.0	688.0	530.0	662.0	501.0
P50	735.0	677.0	756.0	683.0	751.5	680.0
Max	834.0	818.0	850.0	836.0	850.0	836.0
Mecánica						
Mean	746.4	681.3	781.3	680.4	775.4	680.6
Std.Dev.	44.2	57.2	41.8	58.4	44.2	58.1
Min	641.0	549.0	677.0	491.0	641.0	491.0
P50	737.0	684.0	774.0	678.0	770.0	678.0
Max	834.0	797.0	850.0	831.0	850.0	831.0
Minería						
Mean	749.3	688.2	776.2	674.4	771.6	676.7
Std.Dev.	39.9	50.2	42.5	53.8	43.2	53.3
Min	634.0	573.0	673.0	521.0	634.0	521.0
P50	752.0	690.0	774.0	674.5	770.0	675.0
Max	850.0	787.0	850.0	801.0	850.0	801.0
Observaciones	897		3,231		4,128	

4. Results

Estimates

Equation (2) was estimated separately for men and women, considering a total of eight specializations and ten student cohorts, from 2009 to 2018. The estimations and subsequent analyses were carried out using Matlab and Stata. Table 5 reports the estimated parameters for the characteristics that vary across alternatives and for the individual-specific variables, with one coefficient for each specialization. The estimations include cohort fixed effects, with Biotechnology serving as the omitted specialization.

A role model effect is found among female Civil Engineering students at the University of Chile through the presence of female professors, significant at the 10 % level. This indicates that the higher the percentage of female professors a student encounters in common-curriculum courses related to a specialization, the more likely she is to choose that specialization. By contrast, for male students, there appears to be no significant relationship between specialization choice and the percentage of female professors teaching those courses.

In line with findings from Bettinger & Long and Rask & Bailey, this result supports the presence of a female professor role model effect on women's specialization choices within a specific STEM environment.

In contrast, when evaluating whether a role model effect arises through female teaching assistants, the negative coefficient indicates that the higher the proportion of female teaching assistants a student encounters in common-curriculum courses, the less likely she is to enroll in the related specialization.

This result runs counter to the findings of Griffith & Main (2021), who study the effect of teaching assistants on specialization choice in a STEM program. They find that having a female teaching assistant in introductory courses has a positive effect on specialization choice, increasing the likelihood that female students opt for engineering specializations with higher future earnings.

In Griffith & Main's study, teaching assistants are graduates of the program, and may have closer relationships with students due to the course structure. Although course sizes are similar across the two studies (around 100 students on average), teaching assistants within my sample deliver practical sessions on course material, whereas the TAs in Griffith & Main guide students through active-learning projects during classes.

Given the limited information available on teaching assistants in this study, it is not possible to fully understand the mechanism through which female assistants discourage students from entering specializations with a higher presence of female TAs. If they were all graduates or all students from nearby cohorts, it would be easier to interpret how exposure to these female role models affects specialization decisions.

Griffith & Main (2021) can also identify the specialization of each teaching assistant, and they find differences in the effects depending on the assistant's field. In contrast, the dataset used in this study only includes the TAs' gender and the course and section they taught.

Regarding peer effects from female classmates, either from the same cohort or from previous cohorts, no significant relationship is found for either women or men between the probability of choosing a specialization and the percentage of women in common-curriculum courses or in the previous cohort within that specialization.

For female students, both coefficients are positive but not significant. The literature studying same-cohort classmates and specialization choice has found opposite results (Zölitz & Feld, 2020). Meanwhile, Rask & Bailey do find a positive effect from women in previous cohorts.

The coefficient of the *no_plancomun* variable indicates that specializations that offer common-curriculum courses in a given cohort (Computer Science, Mathematics, Industrial, and Chemistry) are more likely to be chosen by both female and male students. The parameter is negative and significant for both groups, with a stronger effect on women's choices than on men's.

The specialization-specific coefficients show that for female students, Industrial, Electrical, Mechanical, and Computer Science are more attractive options than Biotechnology. The remaining specializations do not show significant effects indicating that they are preferred over Biotechnology. For male students, the coefficients are much larger than those for women, and especially for men, all specializations are significantly more preferred than Biotechnology.

Table 5 also shows that having higher PSU Mathematics and Verbal scores affects the probability of choosing each specialization differently. For female students, a higher PSU Mathematics score increases the likelihood of choosing the Computer Science, Electrical, and Mathematics specializations relative to Biotechnology. This effect is largest for the Mathematics specialization, followed by Electrical and Computer Science. For the remaining specializations, there appears to be no significant relationship between having a higher Mathematics score and choosing them over Biotechnology.

The PSU Verbal score appears to be less important in determining female students' specialization choices. Only the Chemistry and Mechanical specializations show a significant effect, which is negative, meaning that higher Verbal scores reduce the likelihood of choosing these specializations relative to Biotechnology. PSU Verbal scores often provide limited information about students' abilities, making these coefficients more difficult to interpret.

For male students, higher Mathematics scores increase the probability of entering the Mathematics, Electrical, and Mechanical specializations. As is to be expected, higher Mathematics scores make it more likely for both women and men to enroll in specializations that require strong mathematical skills, such as Computer Science, Mathematics, and Electrical Engineering. What is particularly interesting is that for women, high Mathematics scores matter more than for men when entering two of the specializations in which they are most underrepresented: Mathematics and Electrical Engineering.

Table 5: Estimates for women and men regressions.

	Mujer		**Hombre**	
	Coef.	Std. Err.	Coef.	Std. Err.
% Profesoras	0.77*	0.44	-0.08	0.25
% Ayudantes	-0.49**	0.24	0.19	0.14
% Compañeras	1.02	1.15	0.65	0.59
% Alumnas gen anterior	0.16	0.66	-0.10	0.33
No Plan Común	-0.73**	0.34	-0.60***	0.16
Computación	1.18*	0.65	3.09***	0.60
Computación × PSU Mate	0.38**	0.19	0.22	0.16
Computación × PSU Leng	-0.01	0.18	-0.09	0.16
Industrial	2.01***	0.68	3.40***	0.61
Industrial × PSU Mate	0.24	0.17	0.12	0.16
Industrial × PSU Leng	-0.24	0.16	-0.39**	0.16
Eléctrica	1.83***	0.68	3.84***	0.61
Eléctrica × PSU Mate	0.62***	0.19	0.32**	0.16
Eléctrica × PSU Leng	-0.24	0.18	-0.27*	0.16
Matemáticas	-1.00	0.69	1.78***	0.60
Matemáticas × PSU Mate	1.05***	0.28	0.80***	0.17
Matemáticas × PSU Leng	0.03	0.27	-0.08	0.17
Química	0.67	0.70	1.39**	0.62
Química × PSU Mate	0.29	0.20	-0.14	0.18
Química × PSU Leng	-0.38**	0.18	-0.34*	0.18
Mecánica	1.66**	0.69	3.39***	0.61
Mecánica × PSU Mate	0.31	0.20	0.29*	0.16
Mecánica × PSU Leng	-0.47**	0.18	-0.46***	0.16
Minería	0.97	0.70	2.66***	0.61
Minería × PSU Mate	0.36	0.23	0.17	0.17
Minería × PSU Leng	-0.35	0.21	-0.56***	0.17
EF Generación	sí		sí	
N. Estudiantes	897		3,231	
Mención omitida: biotecnología				

Table 6 presents the marginal effects of the main variables of interest for the women's and men's models. The values represent how the average probability of choosing each specialization changes when each independent variable increases by one unit—in this case, a one-percentage-point increase.

For women, if the percentage of female professors associated with the Computer Science specialization increases by 1 %, the probability of a student choosing that specialization increases by 0,0818 percentage points. This impact is strongest for Industrial Engineering: a 1 % increase in the percentage of female professors raises the probability of choosing Industrial by 0,1881 percentage points. It is also notable that the effects are larger for Computer Science, Electrical, Chemistry, and Mechanical than for the remaining specializations.

In contrast, if the percentage of female teaching assistants in Computer Science increases by 1 %, the probability that female students choose that specialization decreases by 0,0517 percentage points. The strongest impact again occurs in Industrial Engineering, where the probability decreases by 0,1190 percentage points.

It is important to highlight that these effects are small in terms of probability changes. For exam-

ple, if the percentage of female professors in Computer Science increases from 20 % to 21 %, the probability that a female student chooses that specialization increases only slightly, by about 0,08 percentage points.

Table 6: Average marginal effects by specialization.

	Profesoras		**Ayudantes**		**Compañeras**		**Gen.Anterior**	
	Mujeres	Hombres	Mujeres	Hombres	Mujeres	Hombres	Mujeres	Hombres
Mención								
Computación	0.0818	-0.0106	-0.0517	0.0251	0.1079	0.0851	0.0168	-0.0125
Industrial	0.1881	-0.0169	-0.1190	0.0401	0.2481	0.1359	0.0386	-0.0199
Eléctrica	0.0828	-0.0136	-0.0524	0.0322	0.1092	0.1091	0.0170	-0.0160
Biotecnología	0.0162	-0.0004	-0.0102	0.0009	0.0213	0.0031	0.0033	-0.0005
Matemáticas	0.0186	-0.0050	-0.0118	0.0119	0.0245	0.0404	0.0038	-0.0059
Química	0.0726	-0.0033	-0.0459	0.0079	0.0957	0.0268	0.0149	-0.0039
Mecánica	0.0708	-0.0096	-0.0448	0.0227	0.0933	0.0769	0.0145	-0.0113
Minería	0.0372	-0.0051	-0.0235	0.0120	0.0491	0.0407	0.0076	-0.0060

Model Fit

The model fits the data quite well, Table 7 shows how female students are distributed across specializations in the actual sample and how the model predicts this distribution. For example, for women, the actual proportion entering the Industrial specialization is 43,03 %, while the model predicts that 44,59 % of women will choose that specialization.

For both men and women, the model underestimates the proportion who will choose Biotechnology. The underestimation is small for men (about one percentage point), but for women it underestimates by more than three percentage points. For the remaining specializations, the model slightly overestimates the proportions, but the deviations do not exceed one percentage point, except for Industrial Engineering among women.

In most cases, the ranking of students' preferred specializations is maintained, except for Biotechnology among women, for which the model predicts a much lower proportion, making it the least preferred specialization for them.

Table 7: Predicted and actual major's distribution.

Mujer				**Hombre**			
Rank	Mención	Predicha	Actual	Rank	Mención	Predicha	Actual
1°	Industrial	44.59	43.03	1°	Industrial	30.26	29.99
2°	Eléctrica	12.44	12.04	2°	Eléctrica	21.48	21.29
3°	Computación	12.38	11.82	3°	Computación	16.03	15.85
4°	Química	10.56	10.14	4°	Mecánica	13.8	13.68
5°	Mecánica	10.27	9.92	5°	Matemáticas	6.86	6.81
6°	Minería	5.08	4.91	6°	Minería	6.74	6.69
7°	Matemáticas	2.53	2.45	7°	Química	4.35	4.3
8°	Biotecnología	2.15	5.69	8°	Biotecnología	0.48	1.39

4.1. Counterfactual Analysis for Female Students

To evaluate the potential effects of different scenarios on the distribution of female students across Civil Engineering specializations at the University of Chile, four counterfactual exercises are conducted. These examine increases in exposure to female professors, female teaching assistants, and female classmates. Finally, the effect of higher PSU Mathematics scores for female students is analyzed as well.

Female Professors

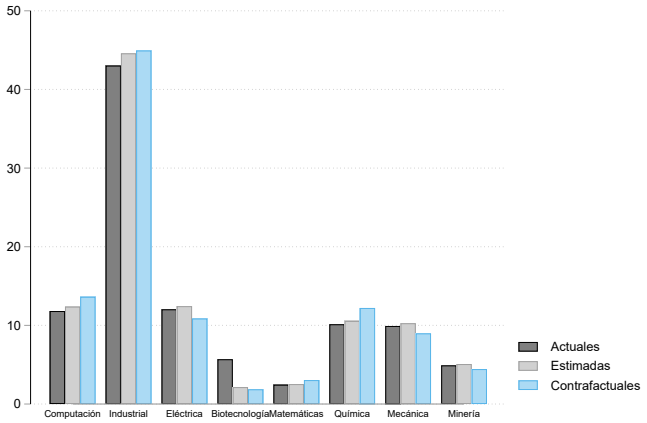
Greater exposure to female professors in common-curriculum courses related to a specialization has a positive effect on women, encouraging them to enter that specialization. As shown in Table 2, the percentage of female professors in common-curriculum courses across specializations and cohorts is always below 50 %, and often zero. Increasing the share of female professors in both common-curriculum courses and specialization-specific courses is a challenge that departments may address through hiring more female faculty.

If in all current common-curriculum courses taken by female students, the share of female professors were increased to 50 %, students would tend to enroll more in those specializations that have such courses[15]. Figure 2 displays the current distribution of women across specializations, the distribution predicted by the model, and the predicted distribution under the counterfactual scenario in which specializations with common-curriculum courses (Mathematics, Chemistry, Industrial, and Computer Science) have 50 % female professors.

I find that a larger number of women choose these specializations, and that under this scenario, the ranking of preferred specializations changes: Computer Science and Chemistry rise in the ranking, making Electrical Engineering the fourth most preferred specialization among female students.

[15]Recall that not all specializations have introductory or related courses taught within the common curriculum.

Figure 2: 50 % of female professors in specializations with introductory courses.



Conversely, if the percentage of female professors is increased for one specialization at a time, it becomes interesting to examine how female students would shift across specializations. In what follows, I conduct this exercise for Mathematics and Computer Science, two specializations in which women are underrepresented and have related courses in the common curriculum.

Female Professors: If the percentage of female professors teaching Mathematics courses is increased to 50 % for all female students, the probability of women choosing the Mathematics specialization increases by 35 %. Since very few women currently choose this specialization, this translates into an increase in the share of women entering Mathematics from 2,53 % to 3,42 % (approximately 8 additional female students).

Figure 3: 50 % of female professors in Mathematics courses.

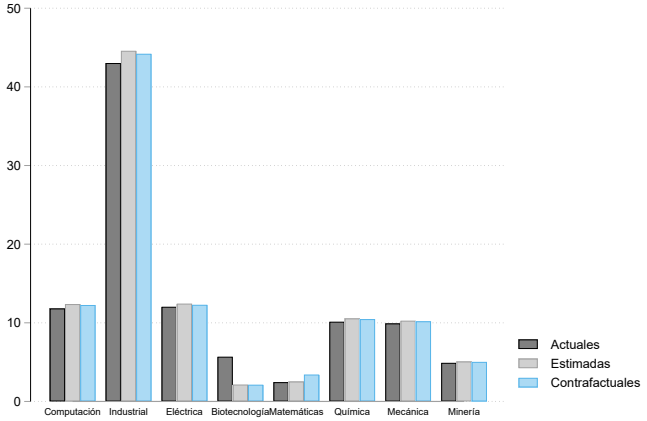


Table 8: Increase of female professors in Mathematics.

Mención	Δ probabilidad	% alumnas contr.
Computación	-0.95	12.26
Industrial	-0.85	44.21
Eléctrica	-1.19	12.29
Biotecnología	-0.81	2.13
Matemáticas	35.26	3.42
Química	-0.86	10.4
Mecánica	-0.86	10.19
Minería	-0.94	5.03

Computer Science: If only the percentage of female professors teaching Computer Science courses is increased, the probability that women choose this specialization increases by 33,8 %. This results in the share of women entering Computer Science increasing from 12,38 % to 16,15 % (equivalent to about 33 additional female students).

Figure 4: 50 % of female professors in Computer Science courses.

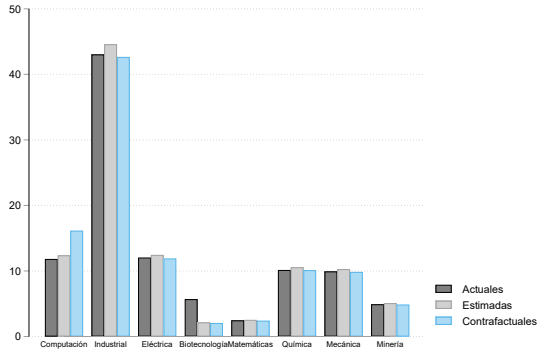


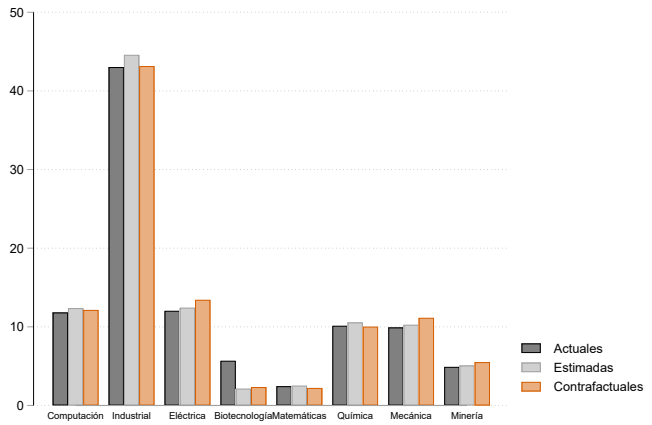
Table 9: Increase of female professors in Computer Science.

Mención	Δ probabilidad	% alumnas contr.
Computación	33,84	16,15
Industrial	-17,38	42,65
Eléctrica	-4,81	11,9
Biotecnología	-0,87	2,05
Matemáticas	-1,04	2,41
Química	-4,02	10,11
Mecánica	-3,79	9,85
Minería	-1,92	4,87

Teaching Assistants

Teaching assistants are typically senior students or recent graduates who have already completed the course and support the lead instructor by teaching practical sessions. As shown in the results, having a higher percentage of female TAs in common-curriculum courses discourages female students from choosing the corresponding specialization.

Figure 5: 50 % of female TAs in specializations with introductory courses.



In this exercise, the percentage of female teaching assistants is increased to 50 % in all specializations with common-curriculum courses. This represents a substantial increase, since the specialization with the highest share of female assistants, Industrial Engineering, has only 27 % on average across cohorts. Figure 5 shows that this discouraging effect leads to increases in the number of women in the Electrical and Mechanical specializations, and small increases also occur in Mining and Biotechnology (all of which lack related common-curriculum courses).

Female Classmates

Since 2014, the Faculty of Physical and Mathematical Sciences at the University of Chile has implemented the Gender Equity Priority Admission Program (PEG[16]), which opened new slots for the first 40 women who remained on the waitlist under the regular PSU admission process. Ten years after the program's introduction, 85 new slots are now offered exclusively to women. Table 10 shows the proportion of men and women in the sample across cohorts. The total number of students increases after the program's implementation, and the share of women rapidly surpasses 20 % in all cohorts where PEG is applied.

[16] Programa Equidad de Género.

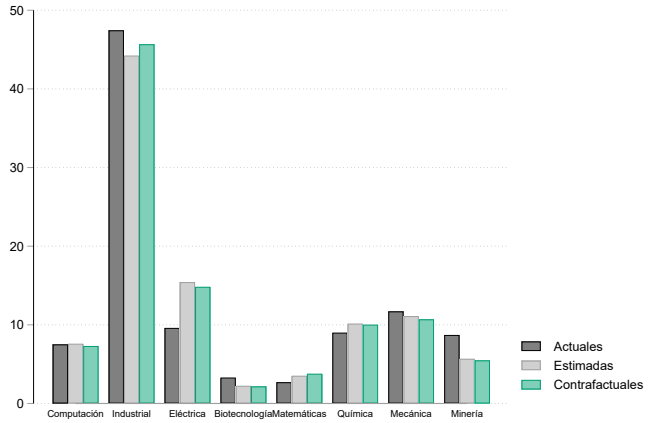
Table 10: Gender distribution by cohort.

Generación	Hombre (%)	Mujer (%)	N total
2009	79.72	20.28	360
2010	85.90	14.10	390
2011	84.45	15.55	373
2012	80.56	19.44	391
2013	83.18	16.82	422
2014	74.78	25.22	448
2015	76.30	23.70	460
2016	74.29	25.71	459
2017	73.74	26.26	438
2018	71.83	28.17	387
Total	78.27	21.73	4,128

Because the sample includes cohorts who entered both before and after the program's introduction, the average percentage of female classmates in common-curriculum courses for post-2014 cohorts (27,4 %) is used, and the percentage of female classmates for pre-PEG cohorts is increased to this value. Figure 6 shows how this adjustment affects the distribution of female students across specializations for cohorts without PEG.

This exercise considers only the courses currently offered in the common curriculum, and the effects vary across specializations. While the proportion of women entering Industrial Engineering and Mathematics increases, the proportion of women entering the remaining specializations decreases.

Figure 6: PEG in cohorts 2009 – 2013



PSU Mathematics Score

Female students entering Civil Engineering demonstrate strong mathematical abilities. Table 4 shows that the median PSU Mathematics score among women in each specialization is around 750 points, with maximum scores reaching 850, the highest possible score, except in the Chemistry and Mechanical specializations.

For the following analysis, it is important to keep in mind that these students already hold very high PSU scores, and increasing scores at the top of the distribution is far more difficult than increasing scores lower down. To account for this, the exercise evaluates how the distribution of women across specializations would change if all female students increased their PSU Mathematics score by 0.25 standard deviations, which corresponds to roughly 10 additional points on the exam.

As shown in Figure 7, the proportion of female students decreases in the Industrial, Biotechnology, Chemistry, and Mechanical specializations. With a higher Mathematics score, some women instead prefer to enter Computer Science, Electrical, and Mathematics, specializations in which women are underrepresented.

Figure 7: Increase of 0,25 standard deviations in PSU Mathematics score.

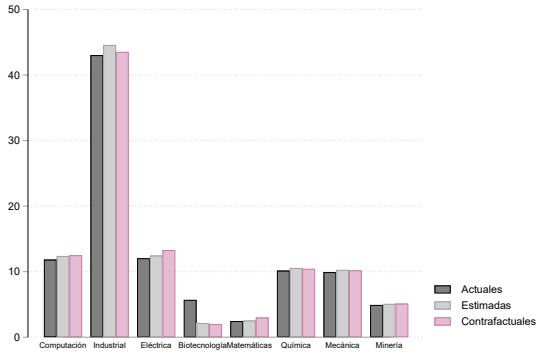


Table 11: Increase of PSU Mathematics score.

Mención	△ probabilidad	% alumnas contr.
Computación	1.00	12.50
Industrial	-2.39	43.53
Eléctrica	6.77	13.28
Biotecnología	-8.07	1.97
Matemáticas	18.26	2.99
Química	-1.24	10.43
Mecánica	-0.78	10.19
Minería	0.44	5.10

PSU Mathematics scores become more relevant when choosing specializations such as Mathematics, Computer Science, and Electrical Engineering. This may be because these fields require stronger mathematical abilities, leading women with lower scores on this exam to enroll less frequently in these majors.

This result is consistent with other studies showing that women require stronger signals of mathematical ability to choose male-dominated specializations (Justman & Méndez, 2018), and that their perception and confidence in their own abilities are lower than those of men, making them less likely to choose these fields (Sikora & Pokropek, 2012b).

These exercises support potential policies aimed at increasing women's participation in specific specializations. For example, increasing the percentage of female professors by hiring more women to teach Mathematics and Computer Science courses in the common curriculum could encourage more female students to enter these specializations. Similarly, interventions that help women feel more capable in mathematics or strengthen their confidence in their mathematical abilities could

also have positive effects.

5. Conclusion

The literature offers several explanations for why women are less likely to choose careers strongly related to science, mathematics, and computer science. One of these explanations points to the influence of role models and the importance of social interactions in shaping students' academic decisions. In this thesis, I study whether exposure to female professors, teaching assistants, and peers during the first years of university affects the specialization choices of female Civil Engineering students at the University of Chile. The aim is to understand whether these role models can encourage women to enter fields in which they are underrepresented.

To analyze this question, I model students' specialization decisions by allowing both individual characteristics and specialization-specific attributes to matter. The empirical strategy includes cohort fixed effects to control for differences across generations of students, such as changes in preferences or in the relative attractiveness of each specialization. The data allow me to distinguish exposure to different types of role models: female professors who teach introductory courses, female teaching assistants (TAs), female classmates, and women from older cohorts. This setting provides a unique opportunity to study the effect of early exposure to women within a common curriculum, prior to the specialization decision.

The results show that female professors play a positive role. Exposure to them increases the likelihood that women choose specializations led by a higher proportion of female faculty, providing evidence of a role-model effect during students' early university years. In contrast, I find that female teaching assistants have the opposite effect, although smaller in magnitude, reducing the probability that women choose specializations where female TAs are more prevalent. On the other hand, I do not find meaningful evidence that peers, whether female classmates in the same cohort or students from previous cohorts, influence women's specialization choices.

These findings suggest several policy implications. Increasing the number of female professors who teach early-year courses, or making sure that students take introductory courses from all specializations before choosing their track, may help female students feel more represented and encourage them to enter fields where women are underrepresented. In addition, the PSU Mathematics score plays an important role, for women, higher scores increase the probability of choosing specializations such as Mathematics and Computer Science. Strengthening mathematics preparation may therefore be an effective tool for reducing gender gaps.

Despite its contributions, this study has limitations. The analysis is based on a single academic program within one university, which may limit the external validity of the findings. Moreover, the analysis focuses on exposure to role models only during the first years of university and does not consider role models from earlier stages, such as parents or school teachers, which previous research has shown to be important. The dataset also includes only passed courses, which prevents a more detailed examination of academic performance as a determinant of specialization choice.

These limitations open up several directions for future research. Better understanding the effect

of female teaching assistants would require information about their academic background, whether they have graduated, and which specialization they completed. Future research could also examine how simultaneous exposure to different types of role models, such as having both a female professor and a female TA, affects specialization choices. Finally, extending this analysis to other engineering programs, business schools, or *bachillerato* programs[17] with common first-year courses could provide a broader perspective on how female role models affect women's academic paths.

[17]These programs consist of a four-semester academic track with a flexible curriculum that includes common core courses as well as elective courses that students can choose according to their area of interest. Graduates can then continue their studies in an undergraduate program at the same university without having to take the PSU again. Universities such as the University of Chile, Pontifical Catholic University of Chile,Pontifical Catholic University of Valparaíso, Universidad del Bío-Bío, Universidad de los Andes, among others, offer *bachillerato* programs.

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